

1 Automated and Weakly-Supervised Dataset Construction Methods

The radar-to-camera projection pipeline described in Section ?? produces bounding box annotations by transforming radar detections into camera pixel coordinates. While this pipeline provides a scalable mechanism for generating training labels without human annotators, the resulting labels are inherently imperfect. Radar detections are subject to angular quantization, range-dependent beam spreading, multipath propagation, sea clutter false alarms, and calibration offsets—each of which introduces spatial error into the projected bounding boxes. The central methodological challenge, therefore, is not merely producing these labels but determining how to use them effectively: how can a camera-based object detector be trained from noisy, radar-generated supervision in a manner that yields performance competitive with models trained on manually curated annotations?

This question sits at the intersection of several active research areas in machine learning, including data programming, semi-supervised learning, teacher-student frameworks, cross-modal supervision, noisy label learning, and knowledge distillation. Each of these fields offers partial solutions, but none directly addresses the specific noise characteristics of radar-projected labels—noise that is systematic rather than random, geometry-dependent rather than class-dependent, and correlated across frames rather than independently distributed. This section surveys each of these research threads, evaluates their applicability to the radar-guided labeling problem, and identifies the theoretical and practical gaps that remain.

1.1 Data Programming and Programmatic Weak Supervision

The data programming paradigm, introduced by Ratner et al. [1], formalizes the idea that domain experts can encode their knowledge as *labeling functions*—simple, heuristic programs that each provide noisy votes on the correct label for a data point. Rather than requiring every labeling function to be individually accurate, the framework models their collective agreements and disagreements through a generative probabilistic model that estimates the true latent labels. Ratner et al. demonstrated that ensembles of twenty or more labeling functions could produce training sets yielding classifiers within a few percentage points of those trained on hand-labeled data, while requiring 2.8 times less labeling time. The key theoretical insight was that correlations among labeling functions—which arise naturally when heuristics share common failure modes—must be explicitly modeled to avoid overconfident label estimates.

The Snorkel system [2] extended data programming into a practical end-to-end framework. Snorkel introduced a noise-aware generative model that jointly estimates labeling function accuracies and correlations, producing probabilistic training labels that can be used directly with any discriminative model. On benchmark tasks spanning text classification, medical record annotation, and image labeling, Snorkel achieved an average improvement of 45.5% over the majority-vote baseline for combining labeling functions, while enabling dataset creation at rates 2.8 times faster than manual annotation. Critically, Snorkel demonstrated that the generative model could recover from systematic biases in individual labeling functions, provided those biases were not perfectly correlated across all functions.

The data programming framework is directly relevant to radar-guided camera labeling.

A radar detection projected into camera coordinates can be viewed as a labeling function that votes on the presence and location of an object in the image. Additional labeling functions could include temporal consistency checks (does the detection persist across frames?), size plausibility constraints (is the projected bounding box consistent with expected vessel dimensions at the estimated range?), and clutter likelihood estimates (does the radar return exhibit characteristics of sea clutter?). The Snorkel framework provides a principled mechanism for combining these heterogeneous signals. However, a significant limitation is that standard data programming treats each data point as receiving independent label votes, whereas radar projections produce spatially structured labels—a calibration offset biases all projections in the same direction, not randomly. Extending the generative model to account for such spatially correlated error structures remains an open problem.

1.2 Pseudo-Labeling and Self-Training

Pseudo-labeling, introduced by Lee [3], is among the simplest semi-supervised learning strategies: a model trained on a small labeled set generates predictions on unlabeled data, and the high-confidence predictions are treated as ground truth for subsequent training rounds. Despite its simplicity, pseudo-labeling has proven remarkably effective. Lee demonstrated that converting soft model outputs to hard pseudo-labels above a confidence threshold τ functions as an entropy regularization mechanism, encouraging the model to produce sharper decision boundaries. The method requires no architectural modifications and integrates naturally with standard training pipelines.

Rosenberg et al. [4] provided an early application of self-training to object detection, demonstrating that iterative pseudo-labeling could improve detector performance when only a fraction of training images carried annotations. Their key finding was that self-training benefited from conservative confidence thresholds—using only the most confident predictions as pseudo-labels reduced the risk of error propagation, at the cost of slower training set expansion. This conservatism-accuracy tradeoff remains a central design consideration in modern self-training systems.

The Noisy Student framework of Xie et al. [5] transformed pseudo-labeling into a state-of-the-art training methodology. The approach trains a teacher model on labeled data, generates pseudo-labels for a large unlabeled corpus, then trains a student model (equal or larger in capacity) on the combined labeled and pseudo-labeled data with aggressive noise injection—including data augmentation, dropout, and stochastic depth. The student becomes the new teacher, and the process iterates. On ImageNet, Noisy Student achieved 88.4% top-1 accuracy, a substantial improvement over the supervised baseline, while also demonstrating dramatic improvements in robustness: the resulting models showed significantly reduced error rates on ImageNet-C (corrupted images), ImageNet-P (perturbation sequences), and ImageNet-A (adversarial examples). The critical insight was that noise injection during student training forced the student to learn representations more robust than those of its teacher, enabling progressive improvement across iterations.

FixMatch [6] unified consistency regularization and pseudo-labeling into a single, elegant framework. The method generates pseudo-labels from weakly augmented unlabeled images, then trains the model to match those pseudo-labels when processing strongly augmented versions of the same images. Only pseudo-labels above a confidence threshold contribute to the loss. On CIFAR-10, FixMatch achieved 94.93% accuracy using only

250 labeled examples—a result that approached the fully supervised baseline with four orders of magnitude fewer labels. The method’s effectiveness stems from exploiting the model’s own consistency as a signal for label quality, an idea with natural extensions to the cross-modal setting.

Caine et al. [7] applied pseudo-labeling to three-dimensional object detection in the autonomous driving domain, using the Waymo Open Dataset. Their work demonstrated that pseudo-labels generated by a high-capacity LiDAR-based detector could be used to train smaller, more efficient models, and that careful handling of confirmation bias—the tendency for pseudo-labels to reinforce the teacher’s errors—was essential for maintaining detection quality across training iterations. This work is particularly relevant to the maritime setting, where a radar-based detection system (analogous to LiDAR in driving) provides initial labels for training a camera-based detector.

For radar-guided camera labeling, the self-training paradigm offers a natural iterative refinement strategy. An initial camera detector can be trained on radar-projected labels, and its high-confidence predictions can then supplement or replace the noisiest radar labels in subsequent training rounds. However, standard pseudo-labeling assumes that the initial labels are noisy but unbiased—that errors are approximately symmetric around the true label. Radar projection errors are not symmetric: calibration offsets produce consistent directional biases, and range-dependent beam spreading causes systematic box size errors that grow with distance. Applying naive self-training to such systematically biased initial labels risks amplifying, rather than correcting, these errors.

1.3 Teacher-Student Frameworks for Semi-Supervised Detection

Teacher-student frameworks extend the pseudo-labeling paradigm by maintaining two coupled models: a teacher that generates pseudo-labels and a student that learns from them. The critical design choice is the mechanism by which the teacher is updated, which determines the stability and quality of the pseudo-label signal.

The Mean Teacher framework of Tarvainen and Valpola [8] introduced the exponential moving average (EMA) mechanism for teacher updates. Rather than copying the student’s weights directly, the teacher’s parameters θ_T are updated as a running average: $\theta_T \leftarrow \alpha\theta_T + (1 - \alpha)\theta_S$, where θ_S are the student parameters and α is a smoothing coefficient (typically 0.999). This averaging produces a teacher that is more stable than the student at any individual training step, yielding pseudo-labels with lower variance. On semi-supervised image classification benchmarks, Mean Teacher substantially outperformed both the Π -model and temporal ensembling baselines, demonstrating that teacher stability was more important than teacher accuracy for effective semi-supervised learning.

STAC (Self-Training with Augmentation-driven Consistency) [9] adapted the teacher-student paradigm specifically for object detection. The approach first trains a detector on labeled data, generates pseudo-labels for unlabeled images using the trained model, filters these pseudo-labels by confidence, and then retrains the detector on both labeled and pseudo-labeled data with strong augmentation applied to the pseudo-labeled images. On COCO, STAC demonstrated approximately two times the data efficiency of fully supervised training, meaning that a detector trained with STAC on N labeled images and M unlabeled images performed comparably to a supervised detector trained on roughly $2N$ labeled images. This result established that unlabeled data, when properly leveraged through pseudo-labeling, could substitute for expensive manual annotation.

Unbiased Teacher [10] addressed a fundamental limitation of prior teacher-student

methods: the pseudo-labels generated by a teacher trained on limited labeled data are biased toward the class distribution of the labeled set, causing the student to inherit and amplify these biases. The solution was twofold. First, the teacher’s classification head was calibrated using focal loss to reduce over-confidence on frequent classes. Second, the EMA update mechanism was applied throughout training rather than only at initialization, allowing the teacher to continuously benefit from the student’s exposure to unlabeled data. On COCO, Unbiased Teacher achieved a 6.8 mAP improvement over the supervised baseline when using only 1% of available labels, with gains persisting at higher labeling fractions. The debiasing mechanism is particularly relevant when radar pseudo-labels exhibit class imbalance, as may occur when certain vessel types or sizes are more reliably detected by radar than others.

Soft Teacher [11] introduced a soft weighting mechanism for pseudo-labels rather than the hard confidence thresholding used in prior work. Instead of discarding pseudo-labels below a confidence threshold, Soft Teacher assigned each pseudo-label a weight proportional to the teacher’s confidence, allowing uncertain but potentially informative labels to contribute to training with reduced influence. Additionally, the method employed a box jittering strategy to refine pseudo-label localization: multiple jittered versions of each pseudo-box were scored by the teacher, and the resulting scores were used to estimate a reliability weight for the box regression target. On COCO, Soft Teacher achieved state-of-the-art semi-supervised detection performance, demonstrating that soft weighting was strictly superior to hard thresholding for pseudo-label selection.

The teacher-student paradigm is architecturally well-suited to radar-guided camera labeling. The radar projection pipeline serves as an initial “teacher” that provides coarse bounding box annotations, and a camera detector trained on these annotations can serve as a refined teacher for subsequent student models. The EMA mechanism provides a natural smoothing of the noisy radar-derived supervision signal. However, existing teacher-student frameworks assume that pseudo-label noise decreases monotonically as training progresses—the teacher improves over time, producing better pseudo-labels for the student. With radar-projected labels, certain error modes (calibration offsets, range-dependent scaling) are fixed properties of the sensor geometry and cannot be corrected by model training alone. This creates a noise floor below which the teacher-student loop cannot improve without explicit correction of the underlying geometric errors.

1.4 Cross-Modal Supervision

Cross-modal supervision exploits the complementary strengths of different sensor modalities to generate training labels for one modality using detections from another. In the autonomous driving domain, the most common instantiation uses LiDAR point clouds—which provide accurate three-dimensional geometry but are expensive to deploy at scale—to generate training labels for camera-based detectors. The key advantage of cross-modal supervision over standard pseudo-labeling is that the teacher and student operate on fundamentally different data representations, reducing the risk of confirmation bias: a camera detector cannot simply memorize the visual features that triggered a radar detection, because radar and camera respond to different physical phenomena.

Hoffmann et al. [12] developed a differentiable warping module that projects radar features into the camera image plane, enabling end-to-end training of radar-camera fusion networks. While the primary application was fusion rather than label transfer, the differentiable projection mechanism demonstrated that geometric correspondences between

radar and camera could be learned jointly with detection features, providing a pathway for correcting systematic projection errors during training. The approach achieved improved detection performance on the nuScenes dataset, particularly for distant and occluded objects where radar provides range information that is unavailable from monocular images alone.

Kim et al. [13] introduced AISail, which uses Automatic Identification System (AIS) transponder data—rather than radar—as a cross-modal supervision source for maritime camera detectors. AIS provides vessel identity, position, heading, and speed, which can be projected into camera coordinates using known camera calibration parameters. AISail demonstrated that automated AIS-camera annotation could produce training sets sufficient for competitive maritime object detection, achieving detection performance within a few percentage points of models trained on manual annotations. The AIS system provides highly accurate position information (GPS-grade, typically under 10 meters), but covers only vessels equipped with AIS transponders, which excludes small recreational craft, kayaks, debris, and other objects of interest. The radar-guided approach complements AIS by detecting all radar-reflective objects regardless of transponder status.

The cross-modal supervision paradigm highlights an important asymmetry between radar and camera perception. Radar provides geometric supervision: it tells the camera system *where* objects are in physical space. Camera provides appearance supervision: it reveals *what* objects look like and enables fine-grained spatial delineation. This complementarity is analogous to the LiDAR-camera relationship in autonomous driving, but with fundamentally different error characteristics. LiDAR provides centimeter-accurate three-dimensional point clouds; X-band marine radar provides two-dimensional (range, bearing) detections with angular resolution on the order of several degrees and range resolution on the order of meters. Consequently, radar-to-camera projected labels are substantially noisier than LiDAR-to-camera projected labels, necessitating more aggressive noise mitigation strategies. Furthermore, radar detections in the maritime environment are contaminated by sea clutter returns, multipath reflections from the water surface, and sidelobe detections—error sources that have no direct analog in the LiDAR-camera setting.

1.5 Learning with Noisy Labels

The noisy label learning literature provides the theoretical foundations most directly relevant to training camera detectors from radar-generated annotations. This subsection reviews the taxonomy of label noise, the major algorithmic approaches to noise-robust training, and—critically—the ways in which radar projection noise departs from the assumptions underlying existing methods.

1.5.1 Noise Taxonomy

Song et al. [14] provide a comprehensive survey of learning with noisy labels, establishing a taxonomy of noise types. *Symmetric noise* (also called uniform noise) corrupts each label with equal probability to any other class, independent of the true label and the instance. This is the simplest model and the most commonly studied, but it is rarely realistic. *Asymmetric noise* (also called class-dependent noise) models corruption probabilities that depend on the true class but not on the specific instance—for example, “ship” might be more frequently mislabeled as “buoy” than as “bridge.” This model is captured by a noise

transition matrix $T \in \mathbb{R}^{K \times K}$, where $T_{ij} = P(\tilde{y} = j \mid y = i)$ gives the probability that true class i is observed as noisy class j . *Instance-dependent noise* allows the corruption probability to depend on both the true class and the specific input features, modeling situations where some examples are inherently more difficult to label correctly than others. Instance-dependent noise is the most realistic but also the most difficult to model and correct.

1.5.2 Foundational Approaches

Natarajan et al. [15] established the theoretical foundation for learning with noisy labels by deriving unbiased risk estimators for binary classification under symmetric and asymmetric noise. They showed that if the noise rates ρ_+ and ρ_- (the probabilities of positive-to-negative and negative-to-positive label flips, respectively) are known, then the standard empirical risk on noisy labels can be corrected to yield an unbiased estimate of the clean risk. The corrected loss takes the form:

$$\tilde{\ell}(f(x), \tilde{y}) = \frac{(1 - \rho_{-\tilde{y}})\ell(f(x), \tilde{y}) - \rho_{\tilde{y}}\ell(f(x), -\tilde{y})}{1 - \rho_+ - \rho_-} \quad (1)$$

where ℓ is the base loss function and $\tilde{y}$ is the observed noisy label. While this result applies to classification, the insight that noise correction requires knowledge of the noise structure—specifically, the transition probabilities—generalizes to all noisy label settings.

Patrini et al. [16] extended this approach to multi-class deep learning by proposing two loss correction strategies based on the noise transition matrix T . The *forward correction* modifies the network’s output probabilities: $\tilde{p}(y \mid x) = T^\top p(y \mid x)$, accounting for the fact that the observed label distribution is a noisy version of the true distribution. The *backward correction* modifies the loss function directly, scaling the per-class losses by the inverse noise transition matrix T^{-1} . Patrini et al. proposed estimating T from the data itself by examining the network’s predictions on a clean validation set or by using the network’s own confident predictions as proxies for clean labels. On CIFAR-10 and CIFAR-100 with synthetic noise, both correction methods substantially outperformed uncorrected training, with forward correction being more numerically stable.

The Generalized Cross Entropy (GCE) loss of Zhang and Sabuncu [17] provides a noise-robust alternative to the standard cross entropy loss without requiring explicit noise estimation. GCE is defined as:

$$L_q(f(x), y) = \frac{1 - f(x)_y^q}{q} \quad (2)$$

where $f(x)_y$ is the predicted probability for the observed label y and $q \in (0, 1]$ is a hyperparameter that interpolates between the mean absolute error ($q \rightarrow 0$, maximally noise-robust but slow to converge) and the standard cross entropy ($q \rightarrow 1$, fast convergence but noise-sensitive). The theoretical analysis showed that GCE is noise-tolerant under symmetric noise for any $q < 1$, and that moderate values ($q \approx 0.7$) provide a good tradeoff between noise robustness and convergence speed. The appeal of GCE is its simplicity: it requires no noise estimation and can be applied as a drop-in replacement for cross entropy.

1.5.3 Sample Selection and Peer Network Approaches

Co-teaching [18] takes a fundamentally different approach to noisy labels by leveraging the observation that two networks initialized differently will disagree on which examples are

noisy. Co-teaching maintains two networks that are trained simultaneously; in each mini-batch, each network selects the examples with the smallest loss (which are most likely to have clean labels) and feeds them to its peer network for gradient updates. The fraction of examples selected is gradually reduced according to a schedule that estimates the noise rate. On CIFAR-10 with 45% symmetric noise, co-teaching maintained accuracy above 70%, whereas standard training collapsed to near-random performance. The method’s effectiveness stems from the “small-loss” heuristic: deep networks tend to learn clean patterns before memorizing noisy ones, so examples with small loss early in training are disproportionately likely to have correct labels.

1.5.4 Noisy Labels in Object Detection

While most noisy label research focuses on image classification, Chadwick and Newman [19] specifically addressed the problem of training object detectors with noisy annotations. Working in the autonomous driving domain, they demonstrated that detector architectures (Faster R-CNN, SSD) are sensitive to both label noise in class assignments and localization noise in bounding box coordinates. Localization noise proved particularly damaging: even moderate perturbations to bounding box coordinates (5–10% of box dimensions) caused significant mAP degradation. Their proposed solution involved noise-aware loss functions that down-weighted the contribution of likely-noisy annotations, identified by inconsistency between the detector’s predictions and the provided labels.

Li et al. [20] further investigated noisy labels in object detection, demonstrating that the two components of detection loss—classification loss and localization loss—respond differently to label noise. Classification noise can be partially mitigated by standard techniques (label smoothing, mixup), but localization noise requires spatial-aware corrections. They proposed a noise-adaptive training strategy that separately estimates and corrects for classification and localization noise, achieving improved detection performance on datasets with synthetically injected annotation errors.

1.5.5 The Critical Gap: Systematic Geometry-Dependent Noise

The noise models described above—symmetric, asymmetric, and even instance-dependent—share a common assumption that is violated by radar-projected labels: they treat label noise as approximately stationary, uncorrelated across examples, or at most correlated with instance difficulty. Radar projection noise has none of these properties. Instead, it exhibits four distinctive characteristics that distinguish it from the noise regimes studied in the existing literature.

Geometric bias from calibration offsets. Extrinsic calibration errors between the radar and camera coordinate systems produce consistent directional biases in all projected bounding boxes. A one-degree error in the radar-camera azimuthal alignment, for example, shifts every projected bounding box by a number of pixels that varies smoothly with the object’s position in the image. This error is deterministic, not stochastic: the same calibration offset biases every detection in the same direction. Standard noise transition matrices cannot capture this spatial structure because they model class-conditional corruption probabilities, not coordinate-dependent spatial shifts.

Range-dependent scaling. X-band marine radar has an angular resolution on the order of several degrees, determined by the antenna beamwidth. A target that subtends a fraction of one beamwidth at close range may subtend multiple beamwidths at long range, but the bearing quantization remains constant. Consequently, the angular uncertainty in the projected bounding box center scales linearly with range in image coordinates (for small angles), while the bounding box size uncertainty depends on both range and the physical size of the target. At ranges beyond several kilometers, the projected bounding box may be displaced by tens or hundreds of pixels from the true object center. This range dependence means that the noise characteristics vary systematically across the dataset: nearby objects receive relatively accurate labels, while distant objects receive increasingly noisy ones. No existing noise model accounts for this structured, continuously varying noise level.

Multipath and sea clutter correlations. Multipath propagation off the sea surface creates phantom radar returns that appear at predictable locations relative to genuine targets, particularly at low grazing angles. Sea clutter false alarms, while individually stochastic, exhibit spatial and temporal correlations—clutter is concentrated at short ranges, intensifies in higher sea states, and produces returns that persist over multiple radar scans. These correlated false positives generate systematic label errors (bounding boxes where no object exists) that are clustered in specific image regions rather than uniformly distributed. Standard noisy label methods assume that false positives are rare and randomly located; the concentrated, persistent nature of clutter-induced false labels violates this assumption.

Temporal correlation within tracks. Radar track-level processing associates detections across multiple scans into persistent tracks, each characterized by a smoothed trajectory. All detections within a single track share the same sensor biases—the same calibration offset, similar multipath characteristics, and correlated measurement errors from the tracking filter. When these tracked detections are projected into consecutive camera frames, they produce temporally correlated label errors: a bounding box that is shifted left in frame t will also be shifted left in frames $t - 1$ and $t + 1$. Training on such temporally correlated labels without accounting for the correlation risks teaching the detector to reproduce the systematic bias rather than learning the true object locations.

Table 1 summarizes the relationship between standard noise models and the characteristics of radar projection noise. The instance-dependent noise model of Cheng et al. [21] comes closest to capturing the structured nature of radar noise, but even this model treats noise as a function of input features (image content) rather than sensor geometry (range, bearing, calibration state). What is needed is a *geometry-dependent* noise model that parameterizes the noise transition as a function of the radar-camera geometric configuration—the extrinsic calibration, the object’s range and bearing, and the sea state—rather than the visual content of the image.

1.6 Knowledge Distillation

Knowledge distillation, introduced by Hinton et al. [22], provides a framework for transferring learned representations from a large, high-capacity teacher model to a smaller, more efficient student model. The key mechanism is training the student to match the teacher’s *soft* output distributions (softmax probabilities at elevated temperature) rather

Table 1: Comparison of standard label noise models with radar projection noise characteristics. Standard models assume stochastic, instance-level noise, while radar projection noise is deterministic, spatially structured, and temporally correlated.

Noise Property	Symmetric / Asymmetric	Instance-Dependent	Radar Projection
Corruption source	Random flip / class confusion	Instance difficulty	Sensor geometry
Spatial structure	None (i.i.d.)	Feature-dependent	Range/bearing-dependent
Temporal correlation	None	None assumed	Strong (track-level)
Bias direction	Unbiased (symmetric) or class-biased	Sample-dependent	Calibration-determined
Localization effect	Not modeled	Not modeled	Primary error mode
Noise rate variation	Constant or class-constant	Varies per instance	Varies with range

than hard labels. Soft targets carry richer information than one-hot labels because they encode the teacher’s uncertainty and inter-class similarity structure—what Hinton et al. termed “dark knowledge.” On image classification benchmarks, distillation from a large ensemble teacher to a single smaller student recovered most of the ensemble’s performance at a fraction of the computational cost.

Li et al. [23] extended knowledge distillation to the noisy label setting, demonstrating that a student trained to match a noise-aware teacher’s soft outputs could outperform both the teacher and a student trained directly on the noisy hard labels. The soft targets acted as an implicit noise filter: the teacher’s uncertainty on mislabeled examples produced softer output distributions that reduced the student’s exposure to incorrect supervision. The “knowledge distillation from noisy labels” framework showed that the distillation process itself provided noise robustness, even without explicit noise modeling.

In the radar-guided labeling context, knowledge distillation offers a two-stage training strategy. A first-stage model is trained on radar-projected labels, inevitably absorbing some of the systematic noise. This model then serves as a teacher, generating soft pseudo-labels for the same images. A second-stage student model, trained on the soft labels, benefits from the teacher’s implicit filtering of noise: objects that the radar labeled but the teacher learned to doubt receive soft, uncertain labels, while consistently detected objects receive confident labels. This soft-label transfer mechanism is particularly well-suited to handling the spatially varying noise of radar projections, as the teacher naturally learns to be less confident on distant or peripheral objects where radar projection errors are largest.

1.7 Co-Training and Multi-View Learning

The co-training framework of Blum and Mitchell [24] provides a theoretical foundation for learning from multiple complementary views of data. The central theorem states that if two feature sets (views) are conditionally independent given the class label, and each view is individually sufficient for classification, then a small number of labeled examples suffices to learn an accurate classifier by iteratively labeling unlabeled examples with one view and training the other. Blum and Mitchell demonstrated this on web page classification, using

page content and hyperlink structure as two conditionally independent views, achieving significant accuracy improvements from co-training with as few as twelve initial labeled examples.

Radar and camera constitute a natural multi-view pair for maritime object detection. Radar provides geometric features—range, bearing, Doppler velocity, radar cross section—while the camera provides appearance features—texture, color, shape, spatial context. These two feature sets are approximately conditionally independent given the presence and identity of a maritime target: the radar cross section of a vessel depends on its material composition and physical geometry, while its camera appearance depends on lighting, viewing angle, and surface texture. Their failure modes are also complementary: radar detects objects in fog, darkness, and rain but suffers from angular ambiguity and sea clutter; cameras provide precise spatial delineation and object classification but fail in low visibility and lack direct range measurement.

The co-training paradigm suggests an iterative algorithm in which a radar-based detector labels images for training a camera detector, and the camera detector’s confident predictions label data for refining the radar tracker. This bidirectional supervision can progressively improve both modalities. However, the conditional independence assumption is violated when environmental conditions simultaneously affect both sensors—for example, heavy rain degrades both radar clutter characteristics and camera image quality—and the theoretical guarantees of co-training weaken accordingly. Characterizing the degree of view dependence between radar and camera in maritime conditions, and developing co-training variants that are robust to partial view dependence, remain open research questions.

1.8 Active Learning for Selective Verification

Active learning provides a complementary strategy to fully automated labeling by selectively identifying the most informative examples for human annotation. Brust et al. [25] developed active learning strategies specifically for deep object detection, proposing acquisition functions that quantify the informativeness of unlabeled images based on the detector’s uncertainty. Their approach prioritized images containing objects with high classification entropy or large disagreement between multiple detection proposals, demonstrating that actively selected annotation budgets could achieve equivalent detection performance to random selection with substantially fewer labeled images.

In the radar-guided labeling pipeline, active learning serves a verification rather than annotation function. The vast majority of radar-projected labels can be used directly for training, but a small subset—identified by high radar-camera disagreement, anomalous projection geometry, or extreme range—can be routed to human annotators for verification or correction. This targeted human-in-the-loop approach preserves the scalability of automated labeling while providing quality control at the points of greatest uncertainty. The selection criterion for verification candidates should incorporate radar-specific uncertainty estimates: detections at the edge of the radar field of view, single-scan detections without track confirmation, and projections that fall near the image periphery where lens distortion is maximal are all candidates for preferential human review.

1.9 Quality Metrics for Automatically Generated Labels

Evaluating the quality of automatically generated labels requires metrics that capture both the overall fidelity of the label set and the specific error characteristics that affect downstream model training. Standard metrics include intersection over union (IoU) between generated and reference bounding boxes, label precision (the fraction of generated labels that correspond to real objects), label recall (the fraction of real objects that receive generated labels), and the aggregate noise rate (the fraction of labels that are incorrect or significantly mislocalized).

For radar-projected labels, these aggregate metrics are insufficient because they obscure the systematic structure of the noise. A label set with 70% mean IoU may perform very differently depending on whether the 30% error is uniformly distributed or concentrated at long range. Radar-specific quality metrics should therefore include: (1) systematic bias, measured as the mean signed displacement between projected bounding box centers and reference annotations, decomposed into horizontal and vertical components; (2) range-dependent accuracy profiles, reporting IoU and localization error as functions of the radar-estimated target range; (3) false alarm spatial density, characterizing the distribution of radar-generated labels that do not correspond to real objects, particularly in clutter-prone regions; and (4) temporal consistency, measuring the frame-to-frame stability of projected labels for tracked objects. These metrics provide the diagnostic information needed to select appropriate noise mitigation strategies and to predict the downstream impact of label noise on detector training.

1.10 Summary and Research Gaps

This section has surveyed the major methodological paradigms for training models from imperfect supervision: data programming, pseudo-labeling, teacher-student frameworks, cross-modal supervision, noisy label learning, knowledge distillation, co-training, and active learning. Each paradigm offers tools applicable to the radar-guided camera labeling problem, but each also carries assumptions that are violated by the specific noise characteristics of radar-projected labels. The most critical gaps are as follows.

First, **no existing noise model captures geometry-dependent systematic noise**. The noisy label learning literature has progressed from symmetric to asymmetric to instance-dependent noise models, but none of these parameterizes noise as a function of sensor geometry—the calibration state, target range, and bearing that determine the error characteristics of radar-projected labels. Developing a geometry-aware noise transition model that conditions the noise distribution on the radar-camera geometric configuration would provide the theoretical foundation for principled noise correction.

Second, **maritime cross-modal labeling remains largely unexplored**. The AIS system [13] demonstrates the feasibility of cross-modal maritime annotation using AIS data, but radar-to-camera label transfer—which covers a broader class of targets including non-transponder-equipped vessels and floating debris—has not been systematically studied. The noise characteristics of radar-projected labels in the maritime domain (sea clutter, multipath, range-dependent beam spreading) differ fundamentally from those of LiDAR-projected labels in the automotive domain, and maritime-specific noise mitigation strategies are needed.

Third, **the combination of co-training with noisy label learning is an open problem**. Co-training assumes clean labels generated by each view for the other; noisy label learning assumes imperfect labels but typically from a single source. Radar-guided

camera labeling naturally combines both challenges—labels from a complementary view that are also noisy—but no existing framework integrates co-training’s multi-view leverage with noisy label learning’s noise correction in a unified approach.

Fourth, **active learning for selective verification of cross-modal labels is underexplored**. While active learning for object detection is well-studied [25], its application to the specific problem of selecting which automatically generated labels to route for human verification—using sensor-specific uncertainty estimates rather than model-based uncertainty alone—has received little attention. A radar-specific acquisition function that accounts for range-dependent noise, clutter likelihood, and projection geometry could enable highly efficient human oversight of the automated labeling pipeline.

Finally, **the interaction between systematic label noise and self-training dynamics is poorly understood**. Self-training and teacher-student methods assume that iterative refinement reduces noise over time. When the initial labels contain systematic biases (from calibration errors) and spatially correlated errors (from clutter patterns), the dynamics of error propagation through self-training iterations may diverge from the well-studied case of random noise. Theoretical and empirical characterization of this interaction is essential for designing robust training pipelines for radar-guided camera labeling systems.

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